Type MS · Plan or specify a single-factor laboratory method

Answer 1 of 4 · Question 5 part a.i · 5 marks

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

5 (a)	(i)	any five from • place substrate into a test/boiling tube/beakers; • place test tube in a (hot) water bath; • add enzyme to substrate; • measure time taken for reaction to occur; • use same volume/concentration of enzyme/substrate; • repeat for other temperatures;	Allow a temperature value Allow example of a reaction e.g. colour change, amount of glucose produced reject amount	(5)
	(ii)	temperature;		(1)
(b)		• less kinetic energy; • less/fewer collisions (with substrate); • less enzyme-substrate complexes formed/less binding; • lower enzyme activity/rate of reaction;		(4)

Total for question 5 = 10 marks

Type MS · Plan or specify a single-factor laboratory method

Answer 2 of 4 · Question 5 part b · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

5(a)	movement of a substance against a concentration gradient/low to high concentration; using energy/ATP;		**(2)**

Question Number	Answer	Additional guidance	Mark
5(b)	• cut pieces of potato to same size/length/mass; • weigh each piece; • place into different solutions; • leave for at least 30 minutes; • remove and dry; • re-weigh/re-measure length; • determine which concentration has no / least change in mass/length;		**(5)**

Question Number	Answer	Additional guidance	Mark
5(c)(i)	0.10 × 100; 7.90 1.3;	Ecf allow correct answer to incorrect figures used in calculation Allow 1.26 / 1.27 for final correct answer	**(2)**

Type MS · Plan or specify a single-factor laboratory method

Answer 2 of 4 · Question 5 part b · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	initial mass/volume/size varies; greater change in mass/volume if potato is bigger; provides a standardised result; allows comparison;		**(2)**
Question Number	Answer	Additional guidance	Mark
5(c)(iii)	• plot a graph; • concentration vs percentage change; • find (from graph) concentration at which percentage change is zero/lowest change; • this is the concentration of the contents;		**(4)**

Total 15 Marks

Type MS · Plan or specify a single-factor laboratory method

Answer 2 of 4 · Question 5 part b · 5 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Plan or specify a single-factor laboratory method

Answer 3 of 4 · Question 3 part b.i · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Question number					

3	(a)	(i)	surface area = 24 (mm^2);
			volume = 8 (mm^3);
			3:1;
				Full marks for correct final answer	3

		(ii)	C 92;		1
		(iii)	movement from a high concentration to a low concentration;		1
		(iv)	• reduces the time taken to become colourless/rate of diffusion/reaction is faster; • the diffusion distance is shorter/greater area over which diffusion/collisions can occur;		2
	(b)	(i)	• use acid heated to different temperatures/idea of different temperatures; • compare/measure the time taken to become colourless/compare rate of diffusion;		2
		(ii)	Any two from: • the side length of the cube/size of the cube/surface area/volume of cube; • acid concentration; • concentration of alkali/indicator in the cube;	accept volume of acid Allow volume of indicator	2

Total 11 marks

Type MS · Plan or specify a single-factor laboratory method

Answer 3 of 4 · Question 3 part b.i · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan or specify a single-factor laboratory method

Answer 4 of 4 · Question 7 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

7(a)	Any four from: • volume of lactose/sugar solution used; • concentration of lactose/sugar solution; • volume of enzyme solution used; • concentration of enzyme; • type of enzyme used; • how to measure amount/concentration of lactic acid / apparatus to measure amount / concentration of lactic acid;	Allow amount	**(4)**

Question Number	Answer	Additional guidance	Mark
7(b)(i)	25 °C;	accept 20 °C to 45 °C	**(1)**

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	Any two from: • too few results/temperatures investigated / temperatures are too far apart/large gaps between temperatures; • peak/maximum lactic acid production could be below 25°C / could be above 25°C;		**(2)**

Question Number	Answer	Additional guidance	Mark
7(c)	Any three from: • little/no increase in lactic acid concentration / lactic acid concentration would reach a maximum/peak/become constant; • enzyme denatures (due to decrease in pH); • all substrate converted/broken down/used up; • so reaction /slows down/stops;		**(3)**

Type MS · Plan or specify a single-factor laboratory method

Answer 4 of 4 · Question 7 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
7(d)	Any five from: • not possible (to be completely certain); • due to (low) number of pH/temperature values tested/more pH/temperature values need to be tested; • pH5 appears to be optimum pH; • because most sugar broken down/converted at this pH; • optimum pH could be between 5-8/2-5; • optimum temperature not known; • no repeats carried out/no means calculated; • anomalies not identified; • there could be other components in milk that increase enzyme activity/control variables not taken into account;	Allow value stated within ranges given	**(5)**

Total 15 Marks

Type MS · Plan or specify a single-factor laboratory method

Answer 4 of 4 · Question 7 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark